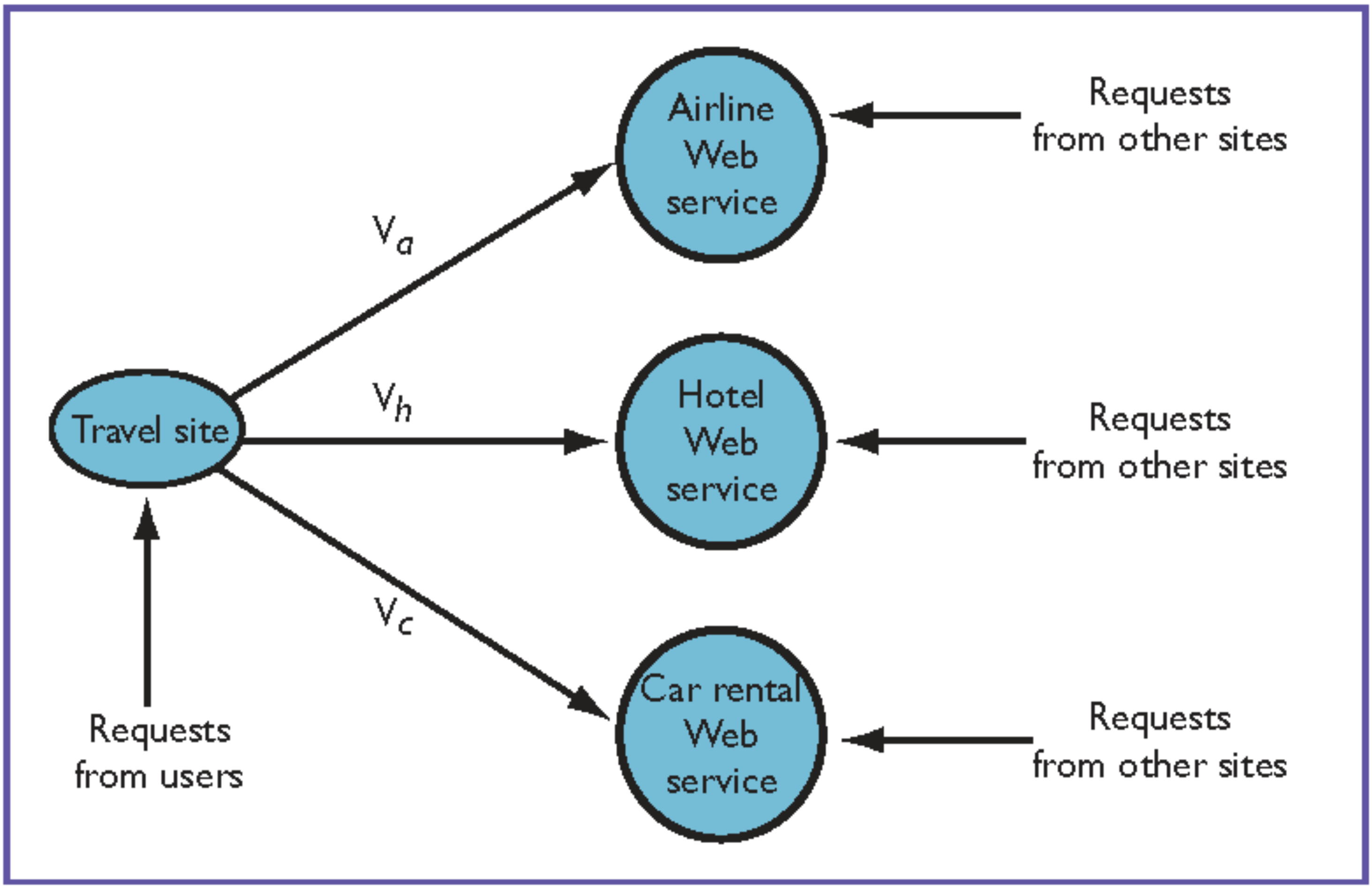


$$X_{TA} = \text{Throughput of travel site}$$

$$X_a = \text{Throughput of airline Web service}$$

$$X_a \geq V_a \times X_{TA}$$

X_a = V_A * X_{TA} + Throughput of the requests from other sites

Scratch: X_a V_A * X_{TA}

$$X_B \quad >= V_{AB} \; X_A$$

$$X_C \quad >= V_{AC} \; X_A$$

Together, they imply:

$$X_A \quad <= \min(X_B / V_{AB} , \; X_C / V_{AC})$$

Scratch:

$$X_A \;\; V_{AB} \;\; X_B \;\; X_B \;\; V_{BD} \;\; X_D$$

$$X_A \; V_{AB} \; <= X_B$$

$$X_A \; V_{AC} \; <= X_C$$

$$X_B \; V_{BE} \; + X_C \; V_{CE} <= X_E$$

$$X_{A} \left(V_{AB} V_{BE} + V_{AC} V_{CE} \right) <= X_E$$

Scratch:

$$X_A \; V_{AB} \; <= X_B$$

$$X_B \; V_{BD} \; <= X_D$$

$$X_A \; V_{AB} V_{BD} \; <= X_D$$

$$X_A <= X_D \; /(V_{AB} V_{BD})$$

Scratch:

Use the table below to work out the response time of the composite web service:

		Web Service 1	
		0.2	**0.3**
Web service 2	**0.2**	0.2	0.3
	0.3	0.3	0.3

E.g. this cell should show the response time of the composite web service when
response time of web services 1 is 0.2
response time of web services 2 is 0.3.

Use the table below to work out the response time of the composite web service:

<table>
<tr><td colspan="2" rowspan="2"></td><td colspan="2">Web Service 1</td></tr>
<tr><td>0.2</td><td>0.3</td></tr>
<tr><td rowspan="2">Web service 2</td><td>0.2</td><td>0.2</td><td>0.3</td></tr>
<tr><td>0.3</td><td>0.3</td><td>0.3</td></tr>
</table>

What is the probability of this occurring? 1/4

Mean response time
= ¼ * (0.2 + 0.3 + 0.3 + 0.3)
= 1.1/4
= 0.275

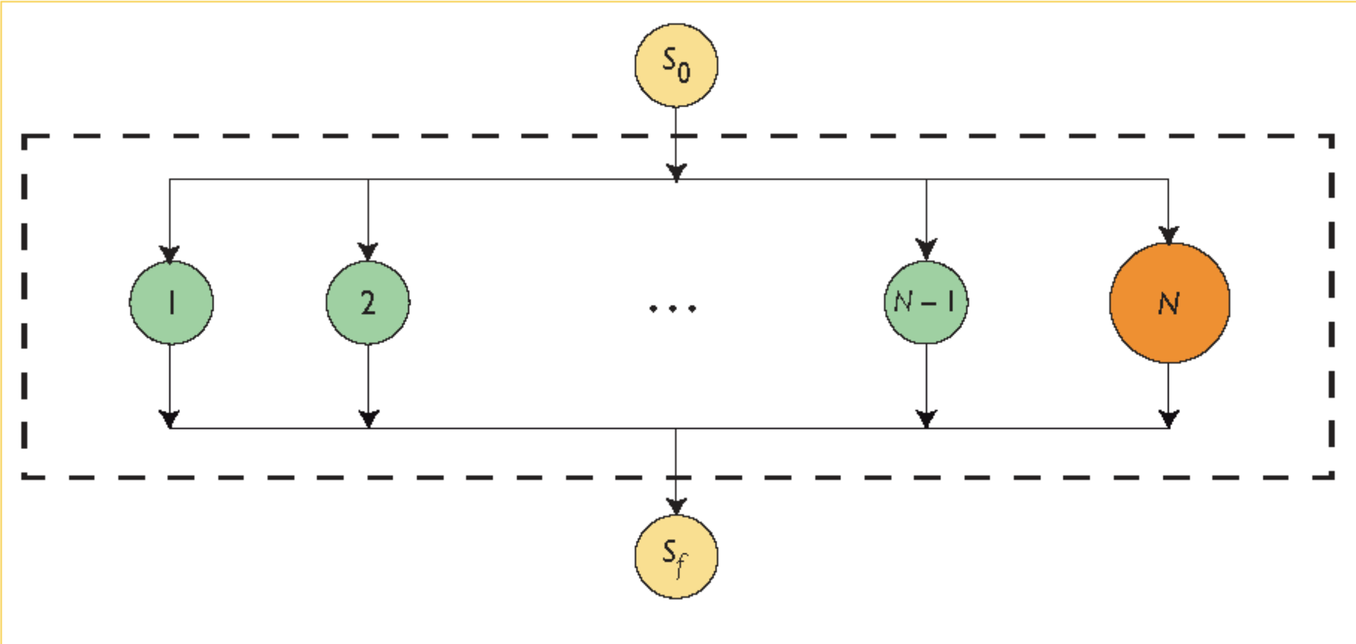
New scenario

		Web Service 1	
		0.2	**0.3**
Web service 2	**0.2**	0.2	0.3
	0.5	0.5	0.5

Mean response time

$= 0.25 * (0.2 + 0.3 + 2 * 0.5)$

$= 0.375$



T($\boldsymbol{\alpha}$) – mean response time

Notation:

$$I = \text{Number of subsystems in the QN}$$
$$S_i = \text{Avg. service time of a station in subsystem } i$$
$$k_i = \# \text{ parallel stations in subsystem } i$$
$$R_i(n) = \text{Response time at subsystem } i \text{ when there're } n \text{ jobs in the QN}$$
$$\bar{n}_i(n) = \text{Avg. } \# \text{ of jobs at subsystem } i \text{ when there're } n \text{ jobs in the QN}$$
$$V_i = \text{Visit ratio of subsystem } i$$

Exercise

- The MVA algorithm on p.36 assumes that you have both visit ratios V_i and mean service time S_i available
- You may recall that service demand $D_i = V_i * S_i$
- Now, let us assume that you are only given the service demands D_i. That is, you know only D_i but you do not know V_i and S_i. How can you modify the MVA algorithm on p.35 so that it can work with knowing service demands only?